Preliminary Results

Code

▶ Code

▶ Code

▶ Code

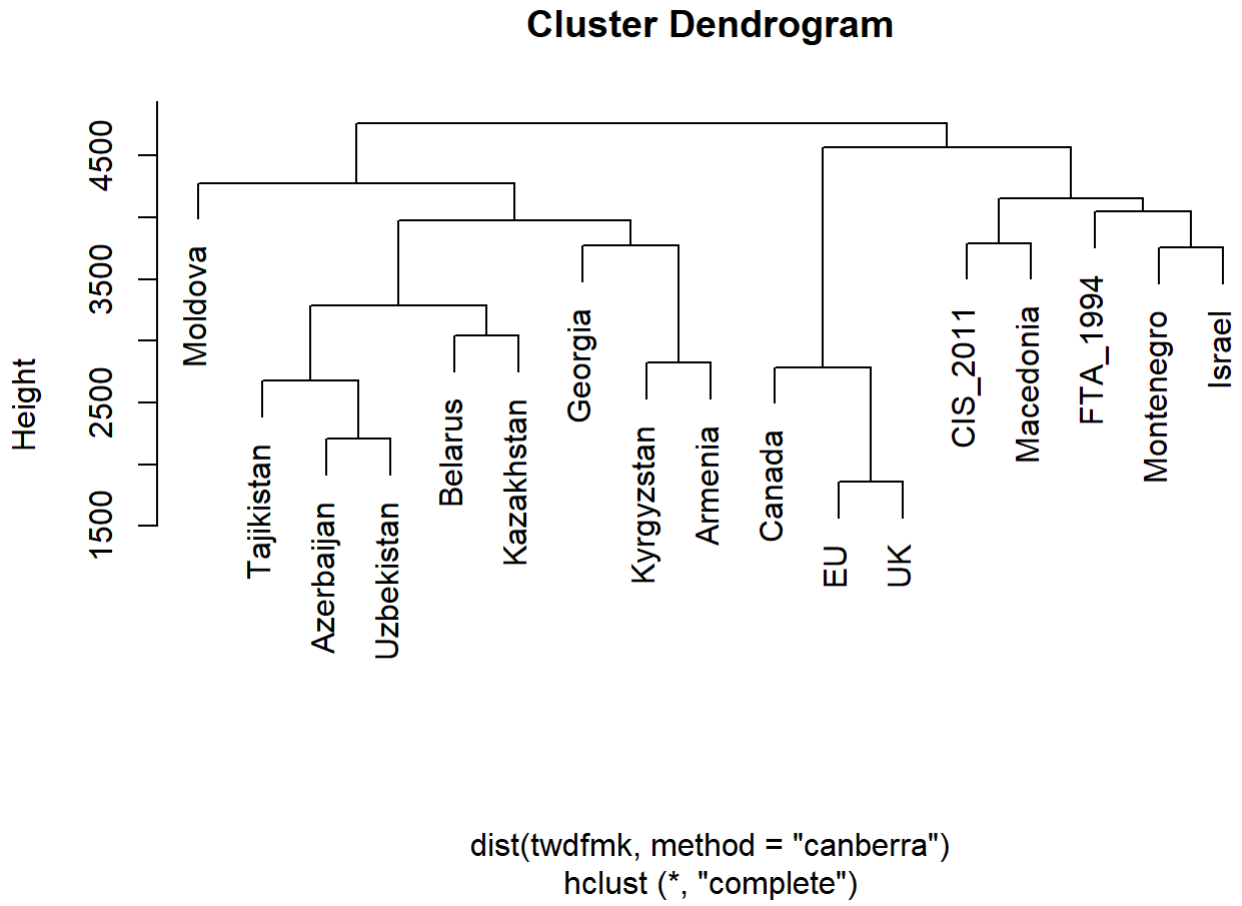
▶ Code

```
Corpus consisting of 17 documents, showing 17 documents:

   Text Types Tokens Sentences  agreement
  text1   548   1761        84 Kyrgyzstan
  text2  2555  18589       608   CIS_2011
  text3  1527  11755       564   FTA_1994
  text4   512   1769        91    Georgia
  text5   540   1515        77    Armenia
  text6  1356   8978       486 Montenegro
  text7  4693  45438      1137  Macedonia
  text8 31651 751209     50889         EU
  text9  3273  32561      1601     Israel
 text10  9297 152343      9111     Canada
 text11   800   3691       131    Moldova
 text12   531   1757        49 Azerbaijan
 text13   675   3929       155    Belarus
 text14   633   2502        71 Kazakhstan
 text15   579   2055        55 Tajikistan
 text16   622   2352        73 Uzbekistan
 text17 10096 167543      7660         UK
```

▶ Code

▶ Code



▶ Code

```
Kyrgyzstan   CIS_2011   FTA_1994    Georgia     Armenia Montenegro  Macedonia
         1          3          1          1           1          3          4
        EU     Israel     Canada    Moldova Azerbaijan    Belarus Kazakhstan
         4          3          3          2           2          2          2
Tajikistan Uzbekistan         UK
         2          2          3
```

▶ Code

▶ Code

```
Running LOOCV loop across 17 documents...
```

▶ Code

```
--- LOOCV Accuracies ---
```

▶ Code

```
Random Forest: 88.24%
```

▶ Code

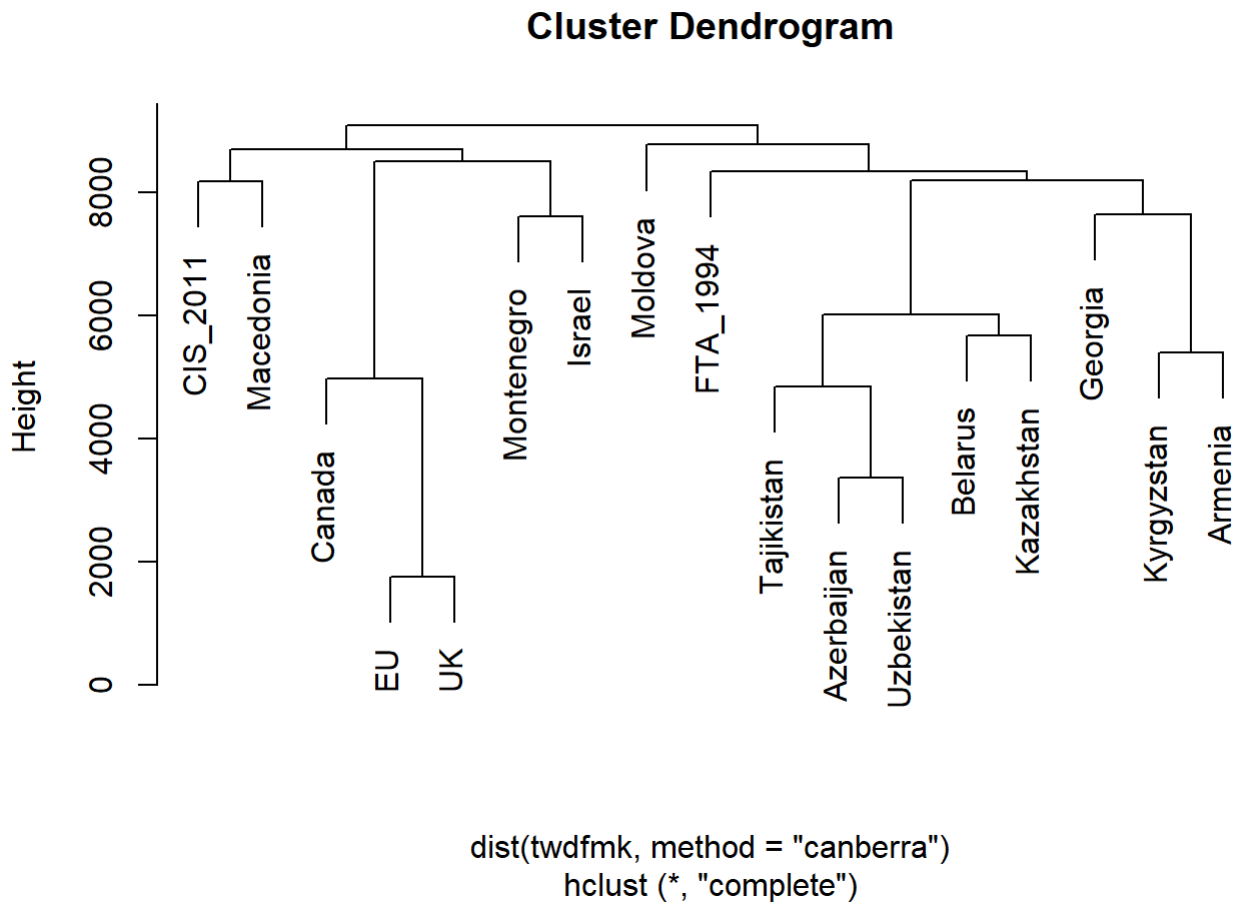
```
Naive Bayes:   88.24%
```

▶ Code

```
Linear SVM:    64.71%
```

▶ Code

▶ Code



▶ Code

```
Kyrgyzstan   CIS_2011   FTA_1994    Georgia    Armenia Montenegro  Macedonia
         1          4          2          2          1          4          2
        EU     Israel     Canada    Moldova Azerbaijan    Belarus Kazakhstan
         4          4          4          2          3          3          3
Tajikistan Uzbekistan         UK
         3          3          4
```

▶ Code

▶ Code

```
Running LOOCV loop across 17 documents...
```

▶ Code

```
--- LOOCV Accuracies ---
```

▶ Code

```
Random Forest: 82.35%
```

▶ Code

```
Naive Bayes:   88.24%
```

▶ Code

```
Linear SVM:    64.71%
```

▶ Code

▶ Code

▶ Code

```
Running LOOCV loop across 17 documents...
```

▶ Code

```
--- LOOCV Accuracies ---
```

▶ Code

```
Random Forest: 76.47%
```

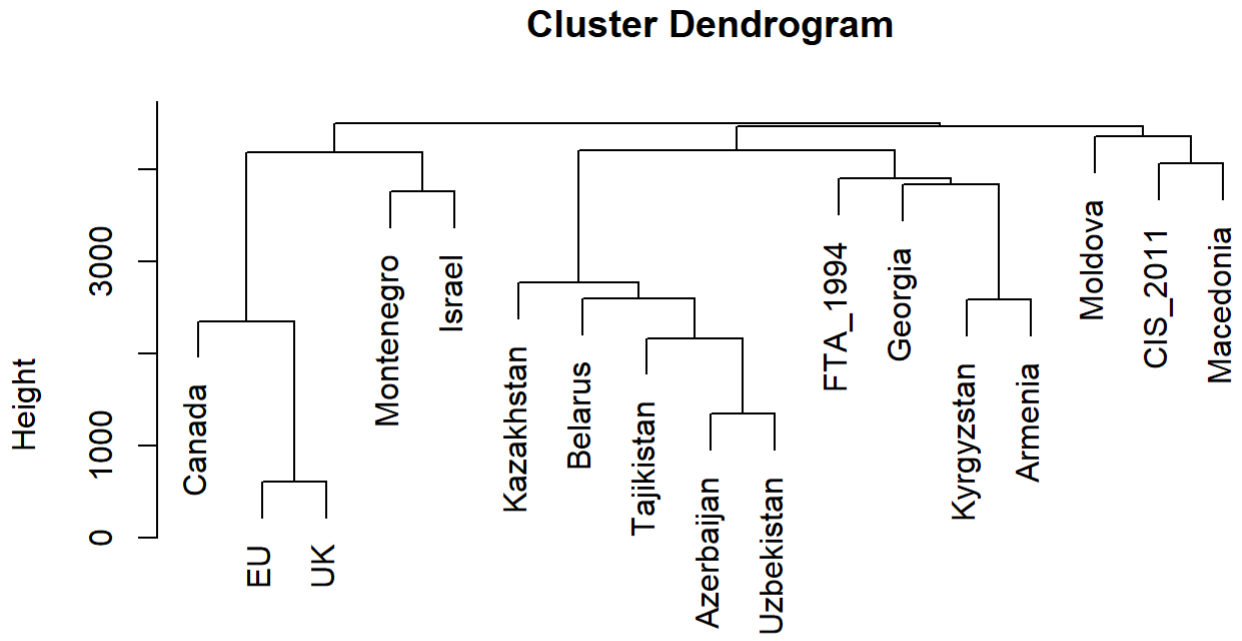
▶ Code

```
Naive Bayes:   88.24%
```

▶ Code

```
Linear SVM:    70.59%
```

▶ Code



▶ Code

```
Kyrgyzstan    CIS_2011    FTA_1994     Georgia     Armenia Montenegro  Macedonia
         3           1           3           3           3           1           4
        EU      Israel      Canada     Moldova Azerbaijan     Belarus Kazakhstan
         4           1           1           2           2           2           2
Tajikistan Uzbekistan          UK
         2           2           4
```

▶ Code

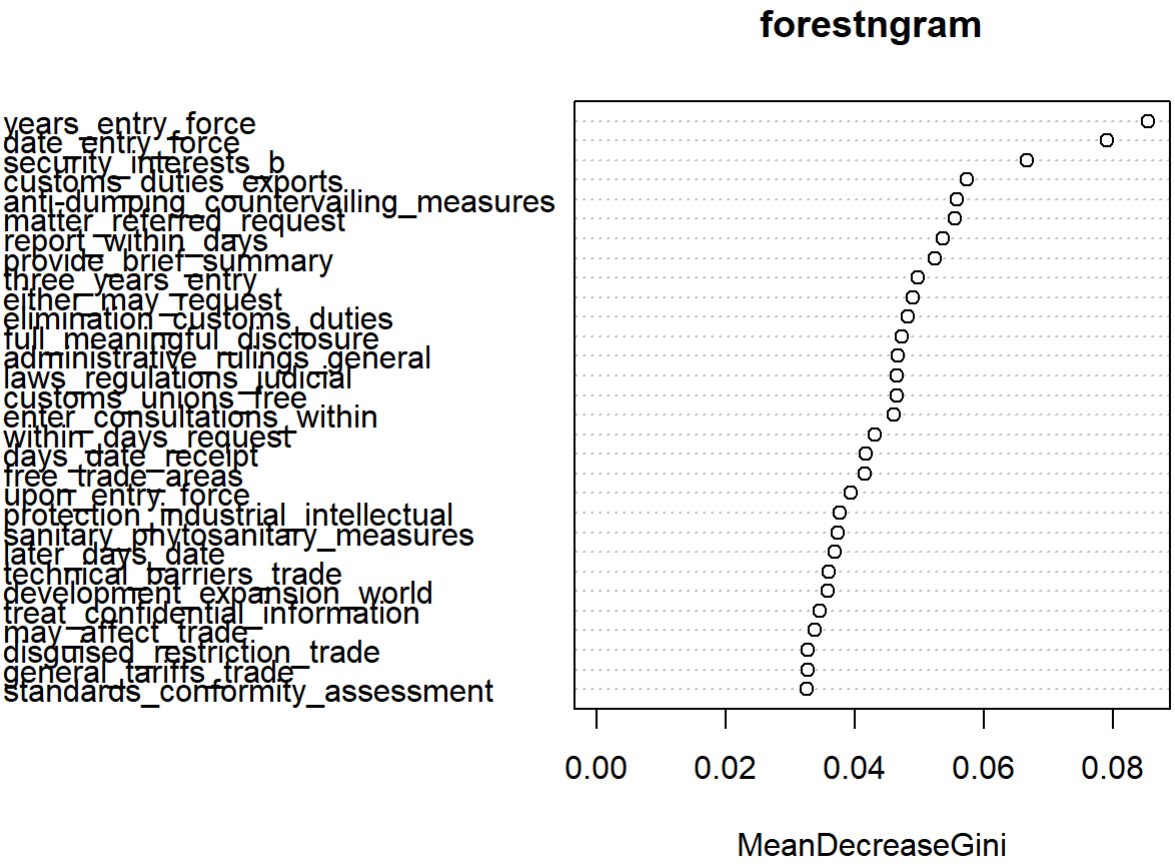
```
                                  1          2 3          4
rate_staging_category         0.001909620 0.00000000 0 0.63228753
elsewhere_specified_included  0.006371645 0.00000000 0 0.16912554
exceeding_exceeding_exceeding 0.003819240 0.00000000 0 0.15628503
originating_third_countries   0.000000000 0.07810841 0 0.14776082
containing_added_sugar        0.016418302 0.00000000 0 0.12631439
containing_less_weight        0.000000000 0.00000000 0 0.09486373
plucked_drawn_without         0.003819240 0.00000000 0 0.08425965
drawn_without_heads           0.003819240 0.00000000 0 0.08425965
intellectual_property_rights  0.033681318 0.03949022 0 0.08075043
added_sugar_sweetening        0.016418302 0.00000000 0 0.08009080
sugar_sweetening_matter       0.016418302 0.00000000 0 0.08009080
movement_certificate_eur      0.007535162 0.00000000 0 0.07917346
years_entry_force             0.019217338 0.00000000 0 0.07404865
necks_hearts_livers           0.003819240 0.00000000 0 0.07222256
take_appropriate_measures     0.009357877 0.00000000 0 0.07129212
```

▶ Code

	1	2	3
ex-works_price_product	0.2100582	0.000000000	0.000000000
provided_value_non-originating	0.2024197	0.000000000	0.000000000
value_non-originating_materials	0.1986005	0.000000000	0.000000000
incorporated_made_part	0.1881611	0.000000000	0.000000000
within_days_date	0.1620959	0.008107864	0.007794562
date_entry_force	0.1386097	0.000000000	0.015589124
information_technical_advice	0.1337600	0.000000000	0.000000000
duties_originating_goods	0.1320203	0.000000000	0.000000000
hs_code_description	0.1246065	0.000000000	0.000000000
customs_duties_originating	0.1224722	0.000000000	0.000000000
hereby_incorporated_made	0.1204163	0.000000000	0.000000000
conformity_assessment_procedures	0.1197685	0.000000000	0.000000000
antidumping_countervailing_measures	0.1193807	0.000000000	0.000000000
ruling_arbitration_panel	0.1180621	0.000000000	0.000000000
days_date_receipt	0.1120808	0.000000000	0.000000000

	4
ex-works_price_product	0.0589508062
provided_value_non-originating	0.0030092732
value_non-originating_materials	0.0091264093
incorporated_made_part	0.0089307934
within_days_date	0.0235382624
date_entry_force	0.0348126899
information_technical_advice	0.0025101878
duties_originating_goods	0.0003107863
hs_code_description	0.0024862904
customs_duties_originating	0.0003107863
hereby_incorporated_made	0.0063692016
conformity_assessment_procedures	0.0052796184
antidumping_countervailing_measures	0.0109288191
ruling_arbitration_panel	0.0180556391
days_date_receipt	0.0128593011

▶ Code



▶ Code

	feature	relative_loglikelihood1
1	rate_staging_category	5.785819
2	ex-works_price_product	4.159686
3	years_entry_force	3.840949
4	value_non-originating_materials	3.260268
5	elsewhere_specified_included	3.250001
6	containing_added_sugar	3.216835
7	provided_value_non-originating	3.172744
8	put_retail_sale	3.106575
9	containing_less_weight	3.096754
10	value_materials_used	3.030840
11	exceeding_exceeding_exceeding	2.992848
12	containing_weight_less	2.689166
13	foodstuffs_human_consumption	2.687965
14	economic_needs_test	2.642383
15	weight_exceeding_kg	2.594624
16	plucked_drawn_without	2.561469
17	drawn_without_heads	2.561469
18	movement_certificate_eur	2.527176
19	materials_heading_exceed	2.491666
20	without_heads_feet	2.416623
21	necks_hearts_livers	2.416623
22	total_value_exceed	2.376878
23	added_sugar_sweetening	2.327586
24	sugar_sweetening_matter	2.327586
25	plates_sheets_strip	2.314131

26	impregnated_coated_covered	2.292310
27	intellectual_property_right	2.292310
28	rectangular_including_square	2.247183
29	plants_plant_products	2.223833
30	provided_total_value	2.200873
31	establishment_arbitration_panel	2.175429
32	hearts_livers_gizzards	2.150320
33	vegetable_fats_oils	2.150320
34	livers_gizzards_known	2.098126
35	air_transport_services	2.071915
36	arbitration_panel_may	2.070970
37	made_available_public	2.048708
38	meat_meat_offal	2.043057
39	extracts_essences_concentrates	2.043057

▶ Code

	feature	relative_loglikelihood1
1	ukraine_government_republic	-4.061536
2	government_republic_belarus	-3.967065
3	authorized_agency_origin	-3.617270
4	regime_free_trade	-3.520470
5	foreign_economic_activity	-3.289898
6	export_import_commodities	-3.236816
7	commodities_originating_third	-3.236816
8	measures_state_regulation	-3.188054
9	commodities_originating_customs	-3.162052
10	customs_territory_another	-3.152364
11	commonwealth_independent_states	-3.133792
12	goods_originating_customs	-3.074650
13	council_heads_cis	-3.074650
14	foreign_economic_relations	-3.074650
15	exporters_importers_carriers	-3.031700
16	intended_customs_territory	-3.015059
17	measures_tariff_nontariff	-2.993324
18	tariff_nontariff_regulation	-2.993324
19	two_valid_copies	-2.947265
20	valid_copies_ukrainian	-2.947265
21	commodities_originating_territory	-2.924510
22	manufacture_commodities_originating	-2.924510
23	council_heads_governments	-2.896925
24	come_within_purview	-2.896925
25	within_purview_present	-2.896925
26	similar_commodities_domestic	-2.896925
27	commodities_domestic_manufacture	-2.896925
28	domestic_manufacture_commodities	-2.896925
29	come_force_date	-2.861823
30	measures_concern_following	-2.861823
31	customs_territory_one	-2.831175
32	originating_customs_territory	-2.826705
33	tariff_non-tariff_regulation	-2.800764
34	beyond_customs_territory	-2.800764
35	hold_regular_consultations	-2.800764
36	_____signature	-2.800764

37	_____signature_signature	-2.800764
38	measures_tariff_non-tariff	-2.799989
39	exceptions_trade_regime	-2.799989
40	russian_languages_texts	-2.799989